

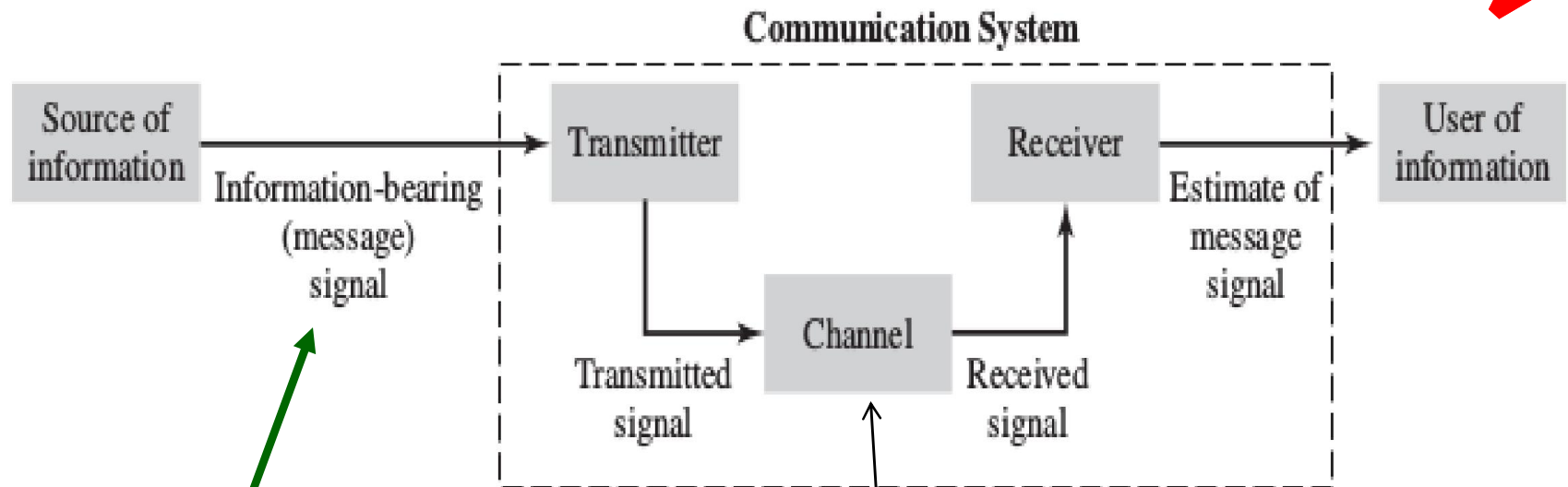
CSE 311:

Data Communication

Instructor:
Dr. Md. Monirul Islam

Components of Communication systems

Review



Most important component

Distortion and Noise

Main challenges

Analog and Digital Signal **Review**

Analog Signal

- Continuous amplitude, i.e., takes any value in a continuous range.
- May be both continuous and discrete time.

Digital Signal

- Discrete amplitude, i.e., amplitude can take only a finite number of values.
- Values need not be always integer.
- Not necessarily always binary, rather M-ary.
- May be both continuous and discrete time.

Benefits of Digital Message/Signal

1. Quality, e.g., enhanced noise immunity
2. Economics

Benefits of Digital Message/Signal

1. noise immunity

- Represented by binary or M-ary pulses

Sent



Benefits of Digital Message/Signal

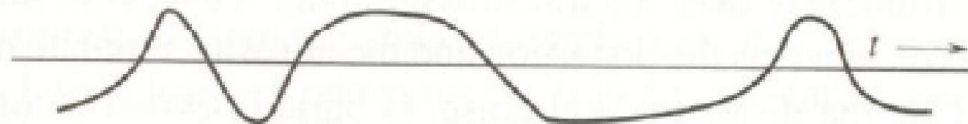
1. noise immunity

- Represented by binary or M-ary pulses

Sent



Received
w/o noise

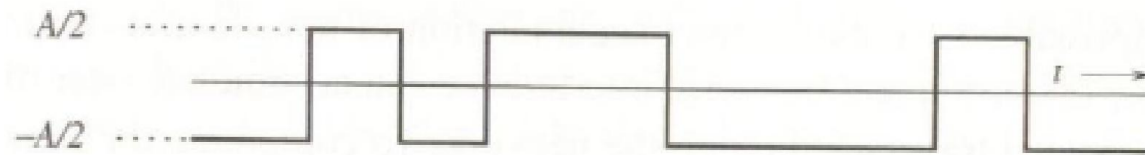


Benefits of Digital Message/Signal

1. noise immunity

- Represented by binary or M-ary pulses

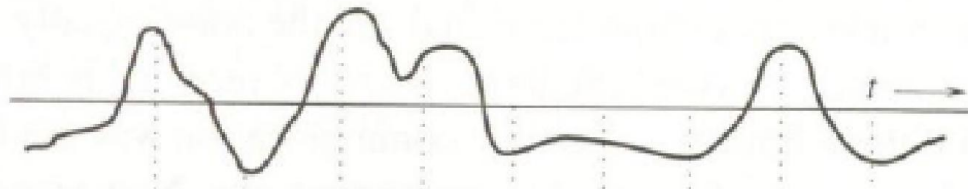
Sent



Received
w/o noise



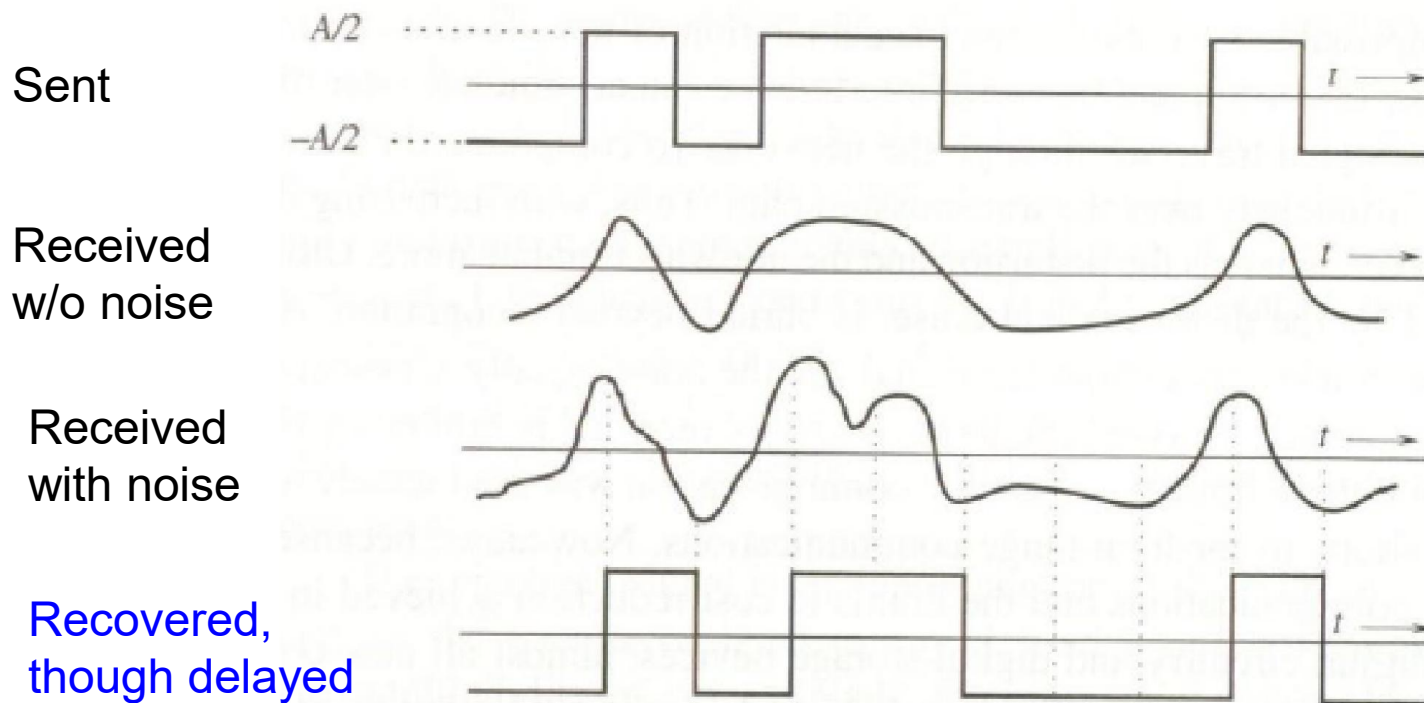
Received
with noise



Benefits of Digital Message/Signal

1. noise immunity

- Recovered despite small distortion and noises

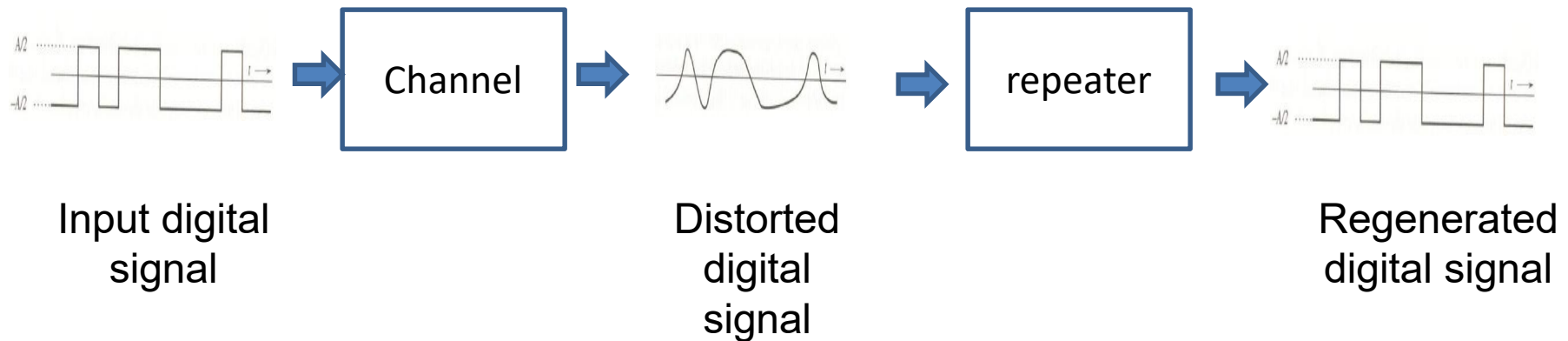


Repeater's Role in Digital Message/Signal

- Distortion and noise are unavoidable in channel
- Repeaters and nodes regenerates digital pulses

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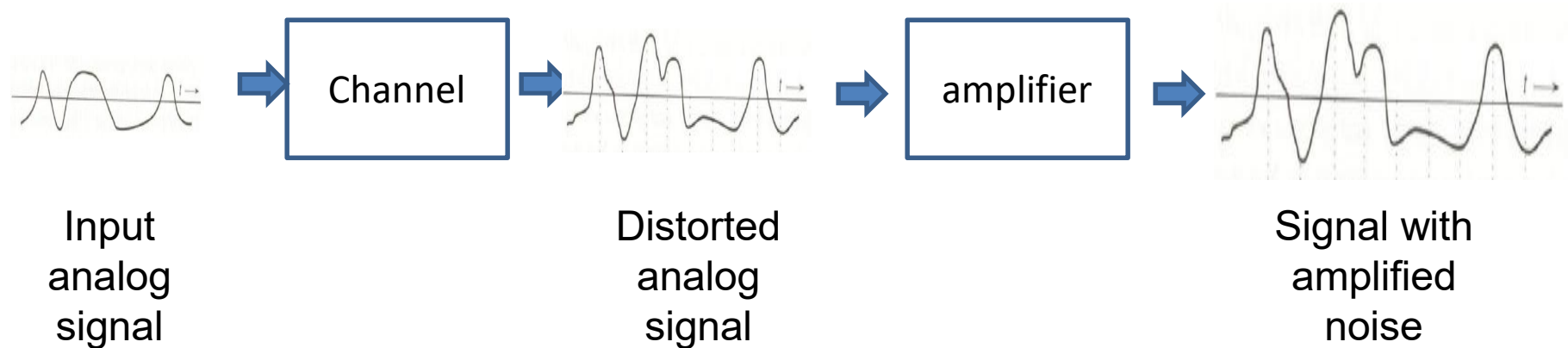


Repeater's Role in Digital Message/Signal

- Distortion and noise are unavoidable in channel
- Repeaters are filters and amplifiers in analog signals
- Amplifier amplifies both signal and noise

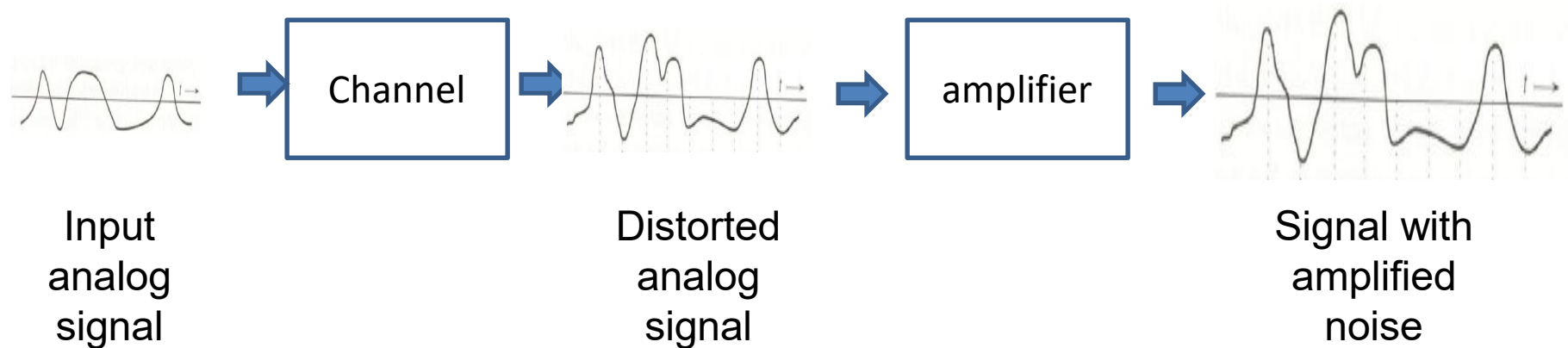
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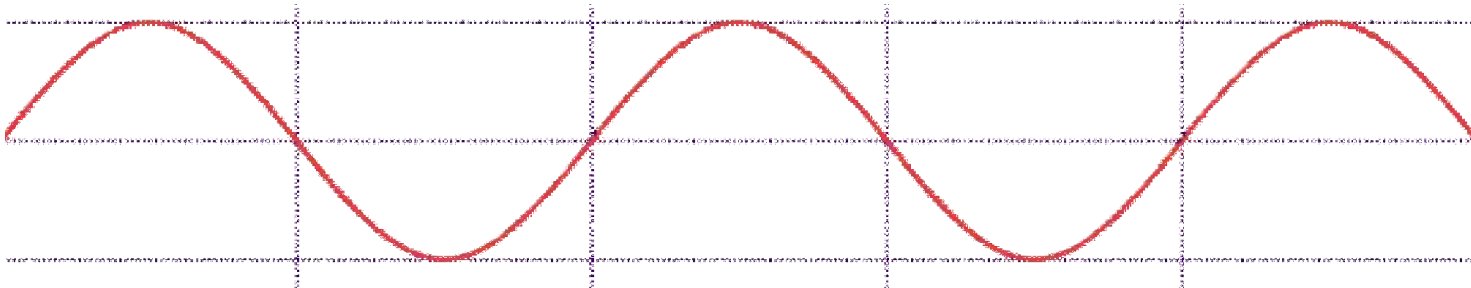
Noise accumulates along the path! No improvement at all !!

Analog to Digital Conversion of Message/Signal

- A/D conversion enables digital communication to convey analog signals

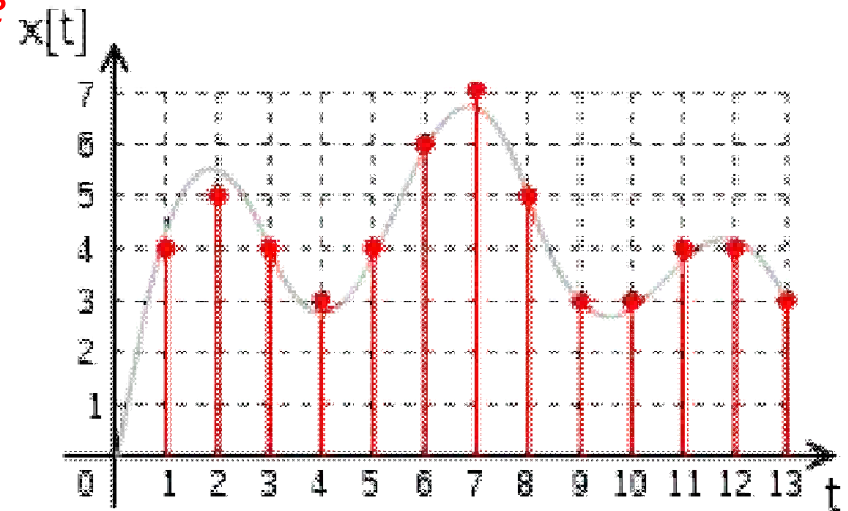
Analog to Digital Conversion of Message/Signal

- A/D conversion enables digital communication to convey analog signals
- Analog signal characteristics
 - values are continuous
 - defined over continuous/discrete time



Analog to Digital Conversion of Message/Signal

- A/D conversion enables digital communication to convey analog signals
- Digital signal characteristic's
 - values are a *finite discrete* set
 - defined over *preferably discrete time*

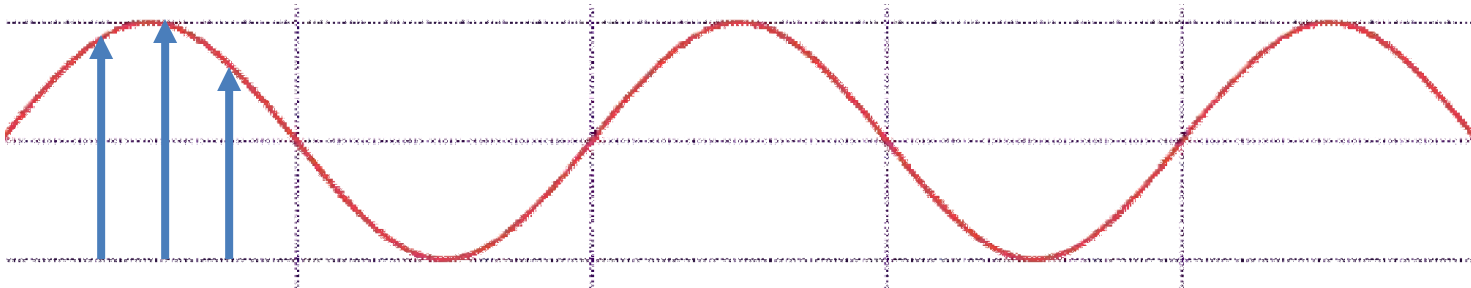


Analog to Digital Conversion of Message/Signal

- 2 major steps
 - Sampling
 - Quantization

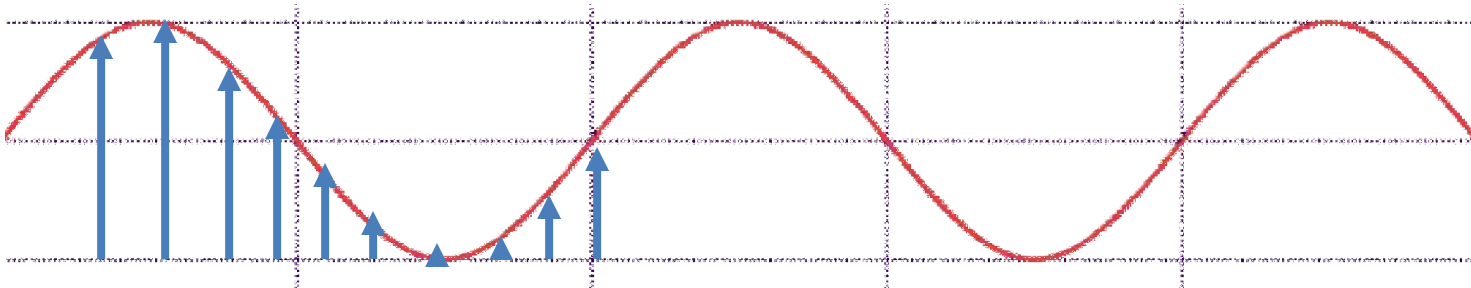
Analog to Digital Conversion of Message/Signal

- Sampling
 - Governed by Nyquist 's Sampling theory
 - Selects points for sampling



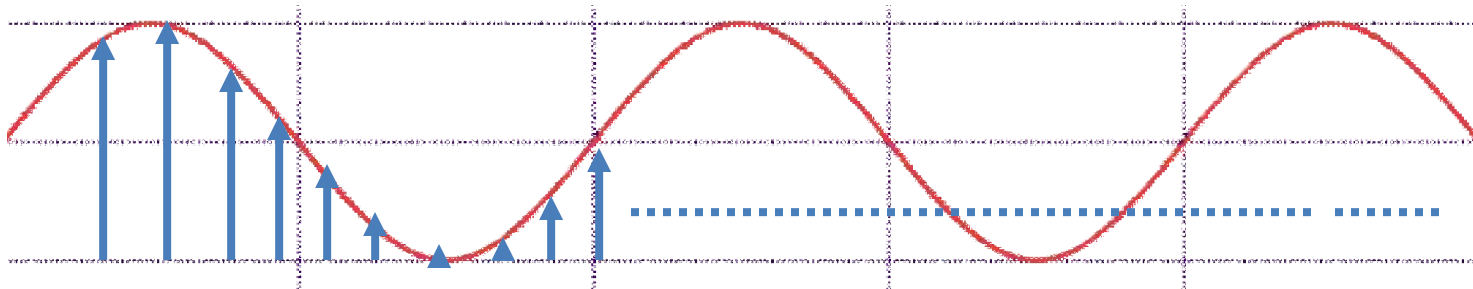
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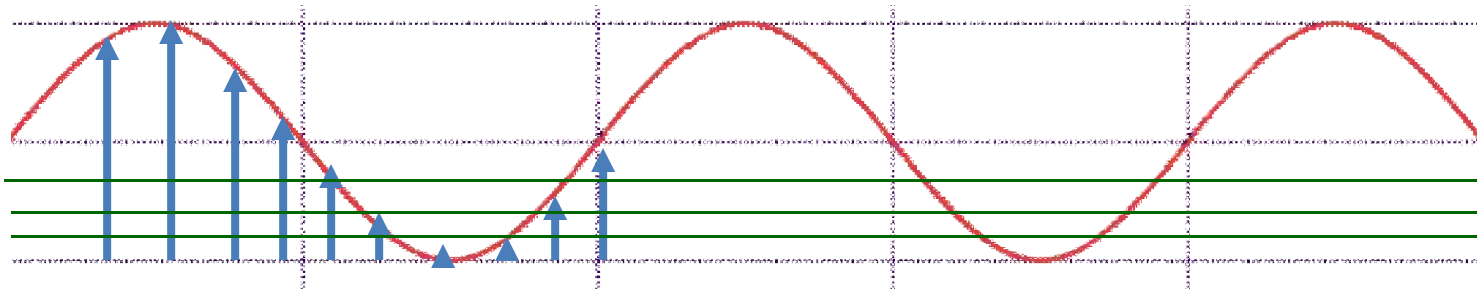
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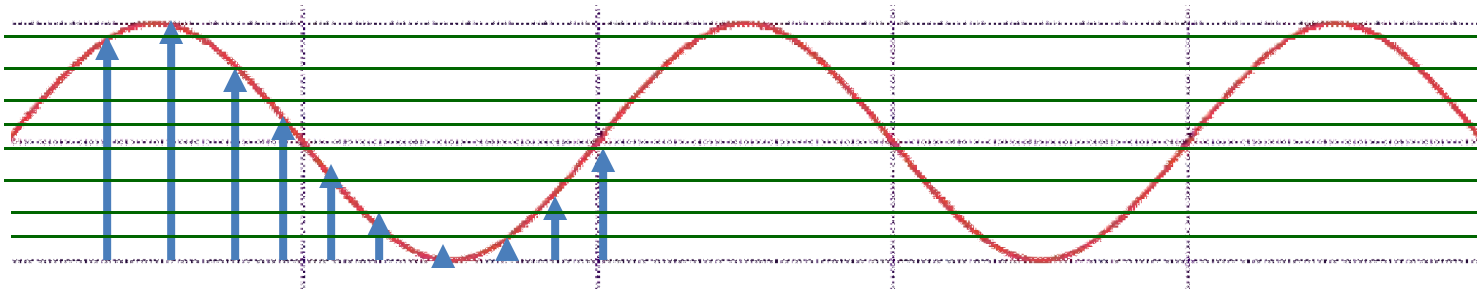
Analog to Digital Conversion of Message/Signal

- Quantization
 - Values replaced by a set of L distinct values
 - Usually $L = 2^k$



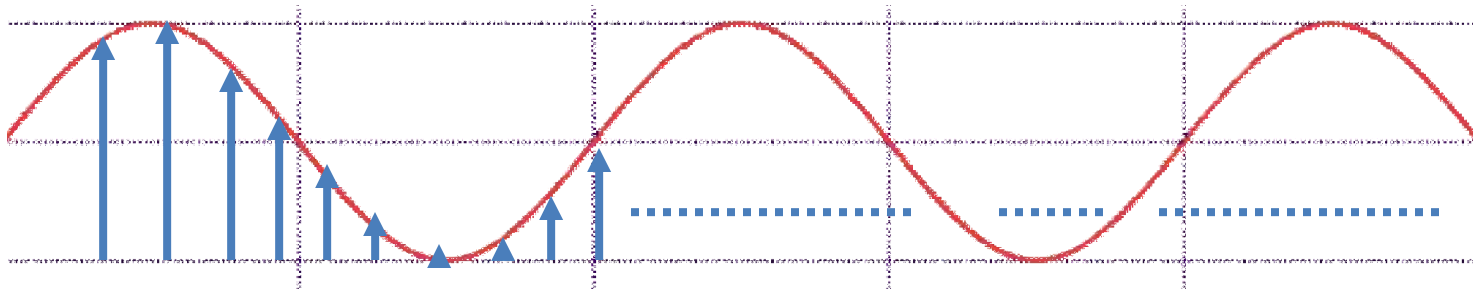
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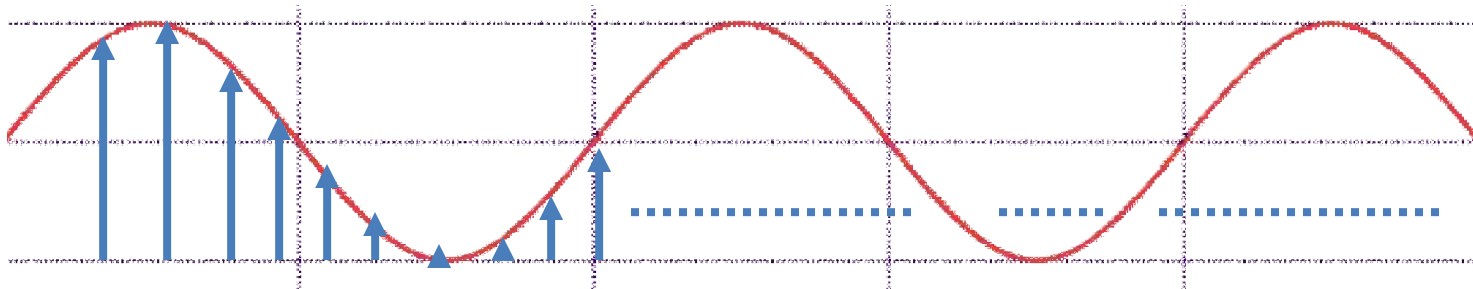
- Increasing sampling rate retains original shape



Analog to Digital Conversion of Message/Signal

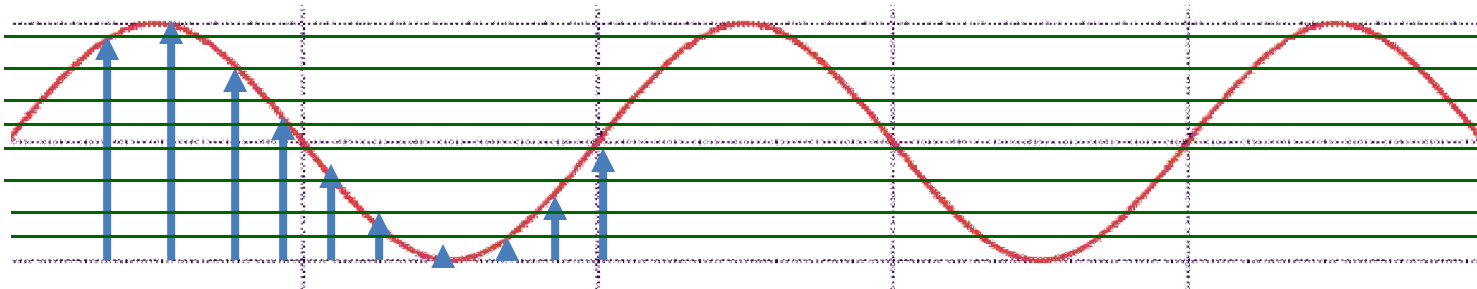
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Remember
Nyquist's theorem!



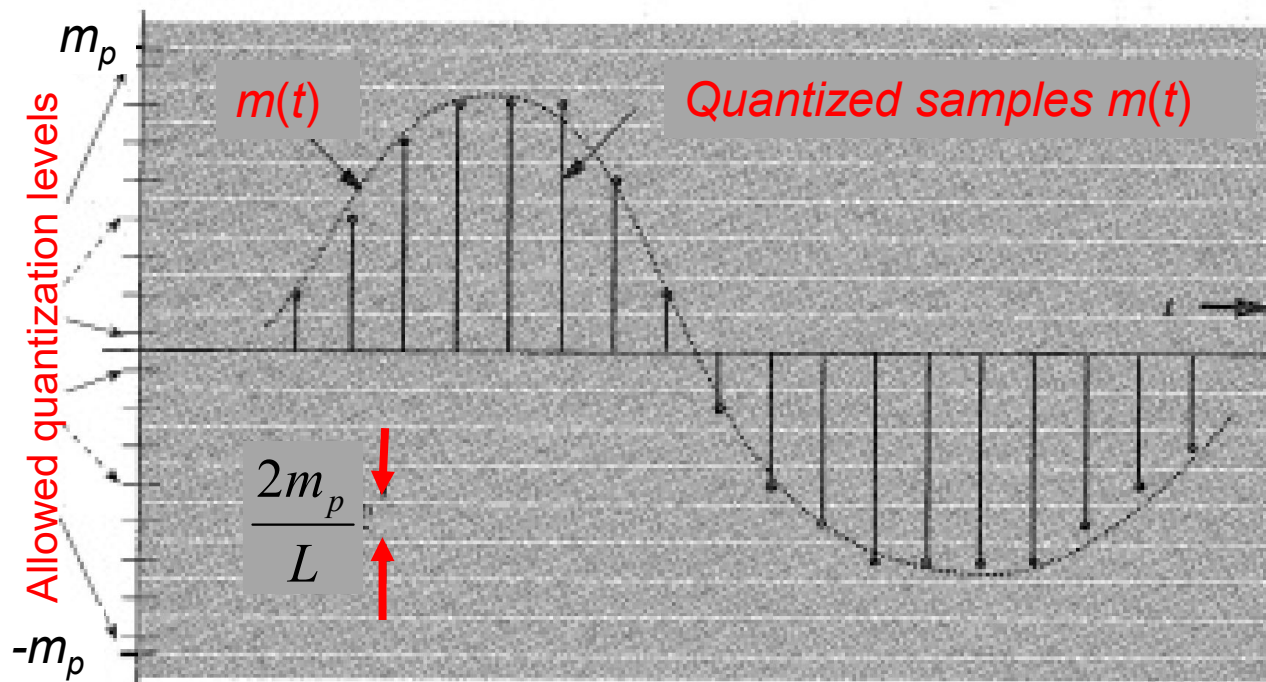
Analog to Digital Conversion of Message/Signal

- Increasing Quantization level L
 - increases accuracy
 - more noise immunity
 - but requires higher channel bandwidth



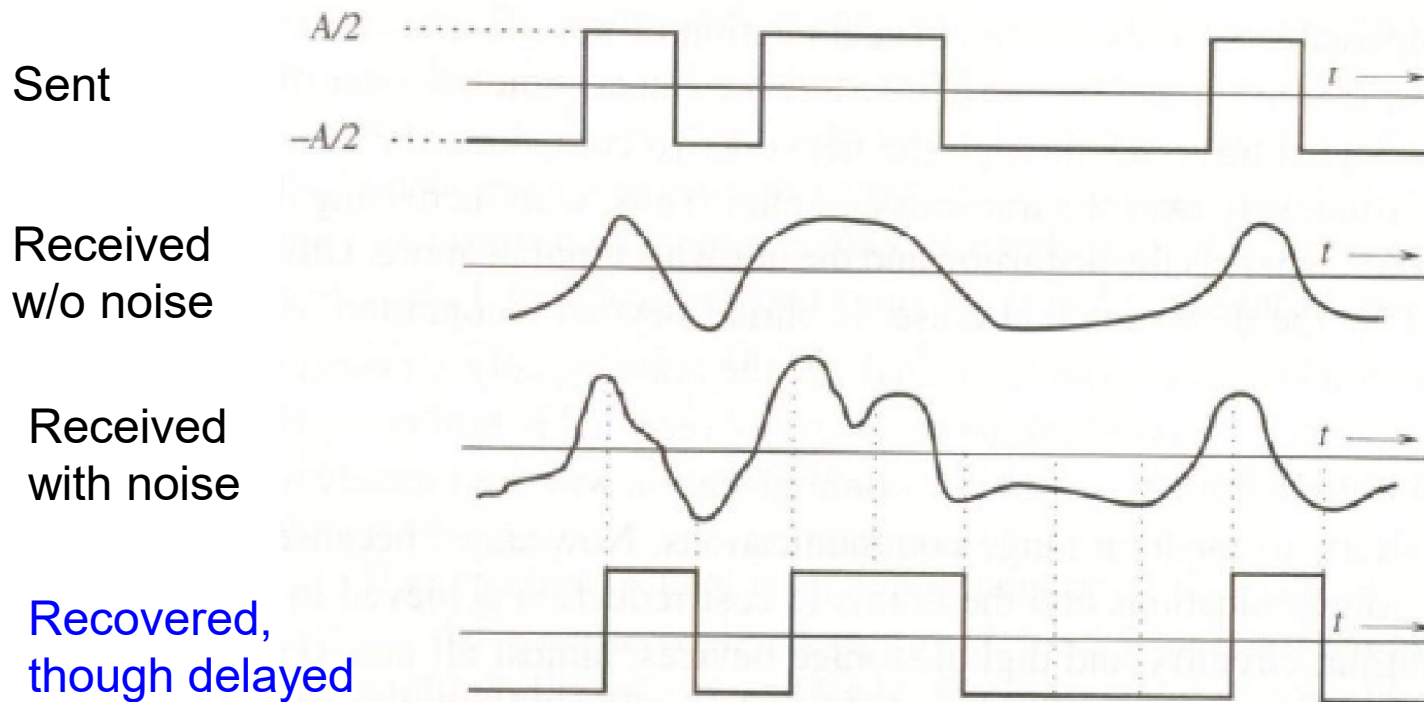
Analog to Digital Conversion of Message/Signal

- Another example of sampling and quantization



Analog to Digital Conversion of Message/Signal

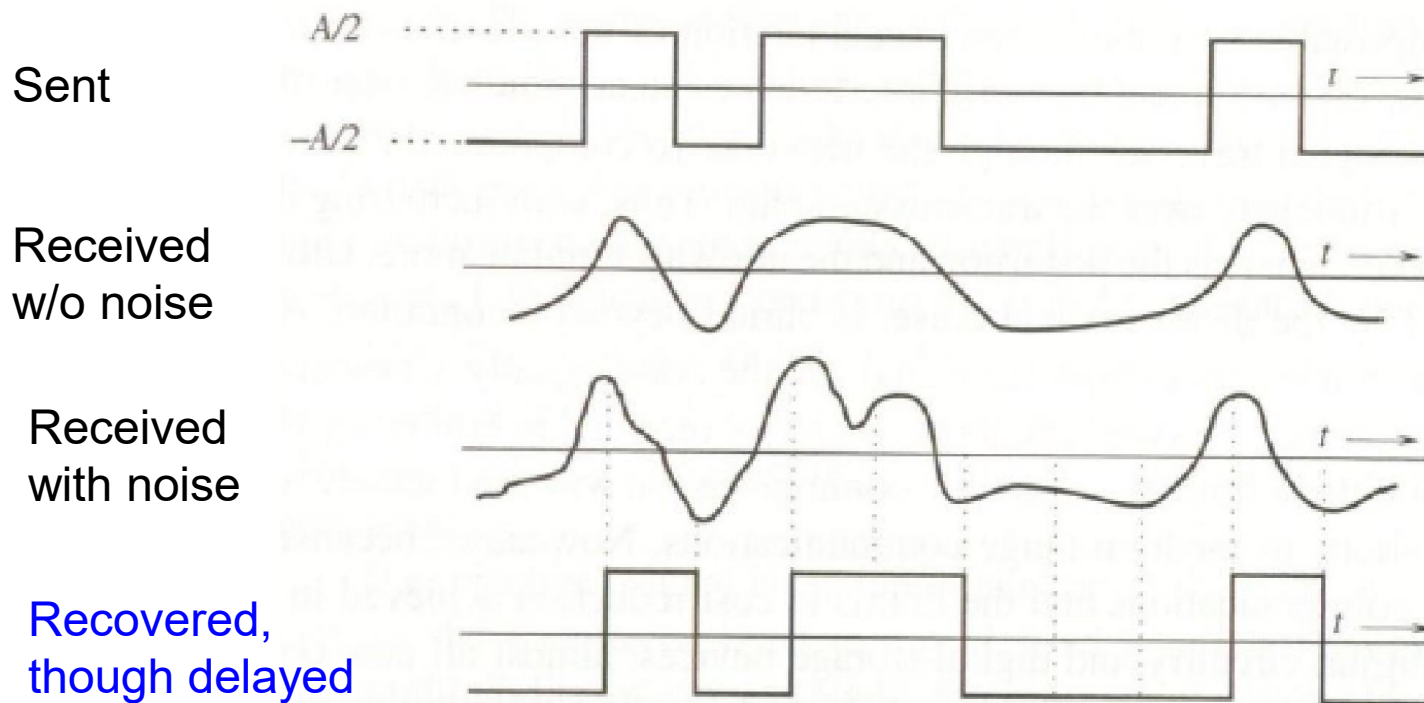
Recall this figure



Analog to Digital Conversion of Message/Signal

- Detection is easy when $A \gg \text{noise}$
- Usually $A \gg 5\text{-}10$ times of noise

Recall this figure



Analog to Digital Conversion of Message/Signal

- Problem: quantization error is unavoidable

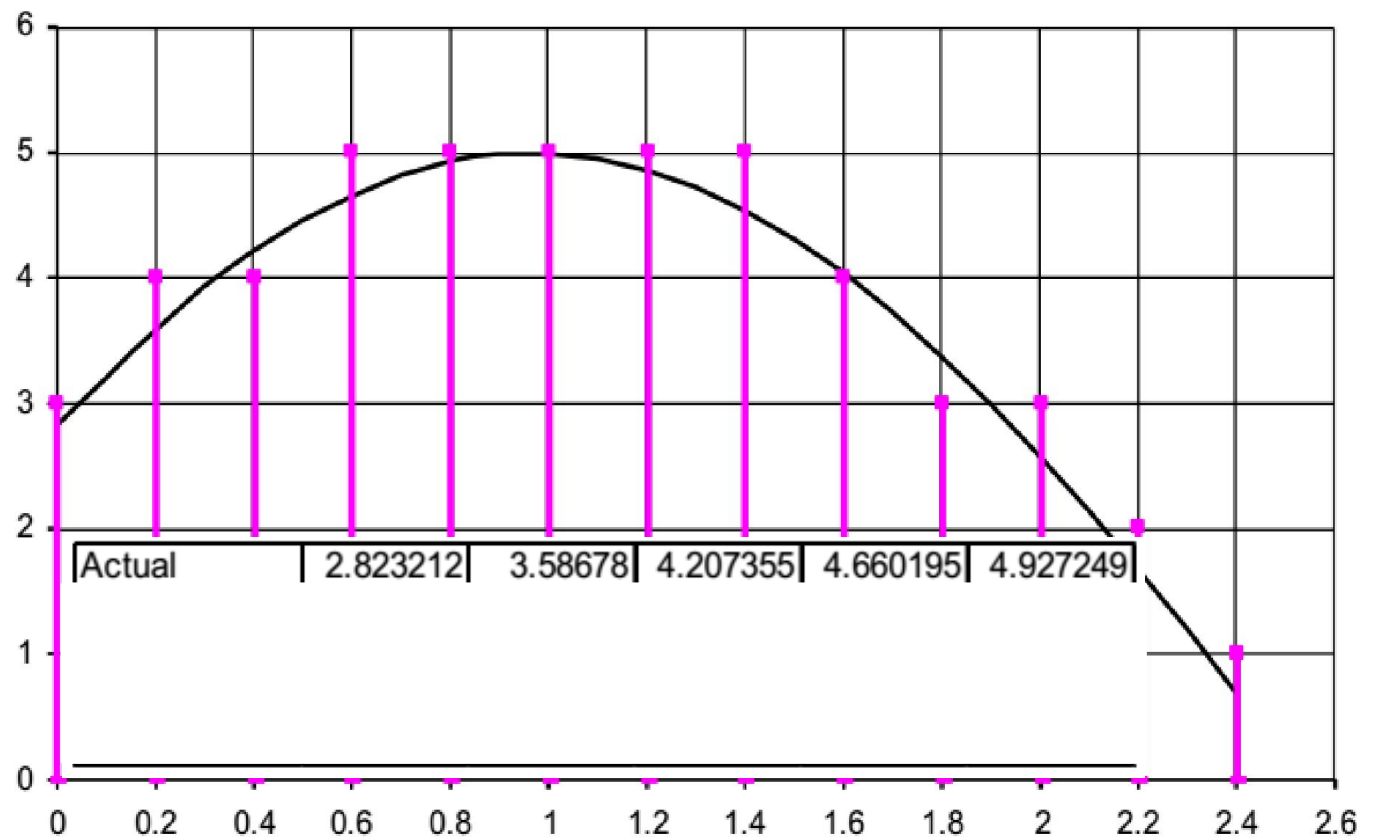
Analog to Digital Conversion of Message/Signal

- Problem: quantization error is unavoidable



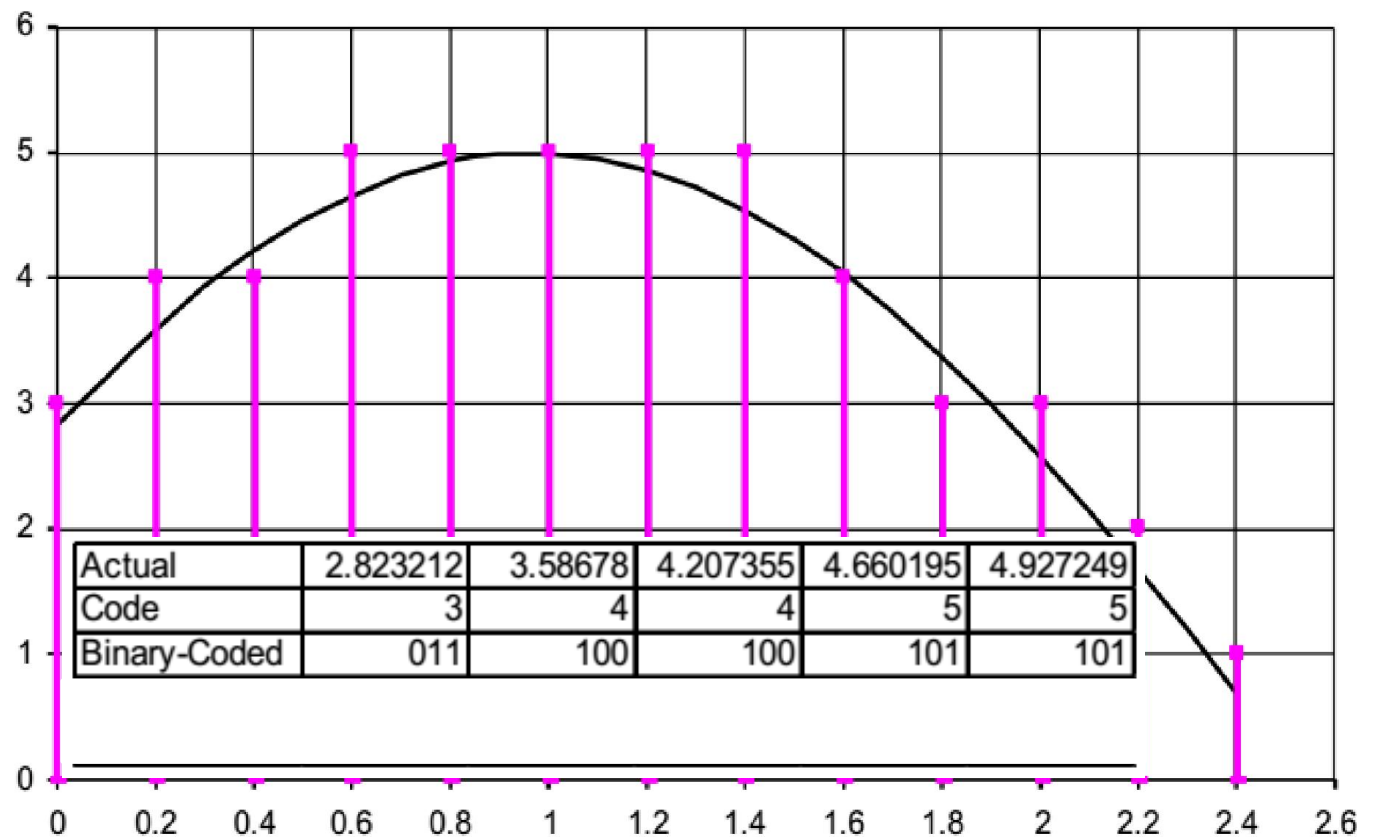
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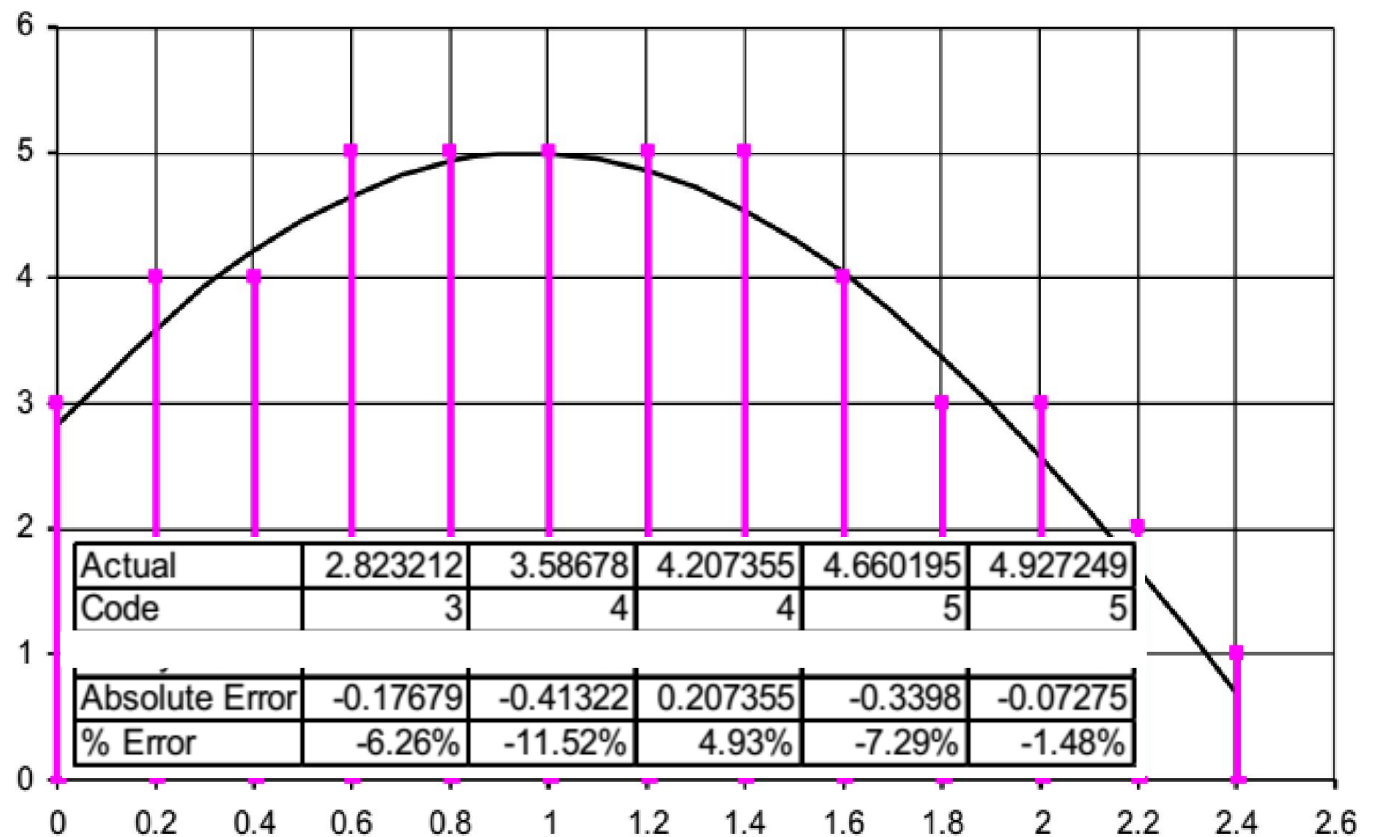
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Analog to Digital Conversion of Message/Signal

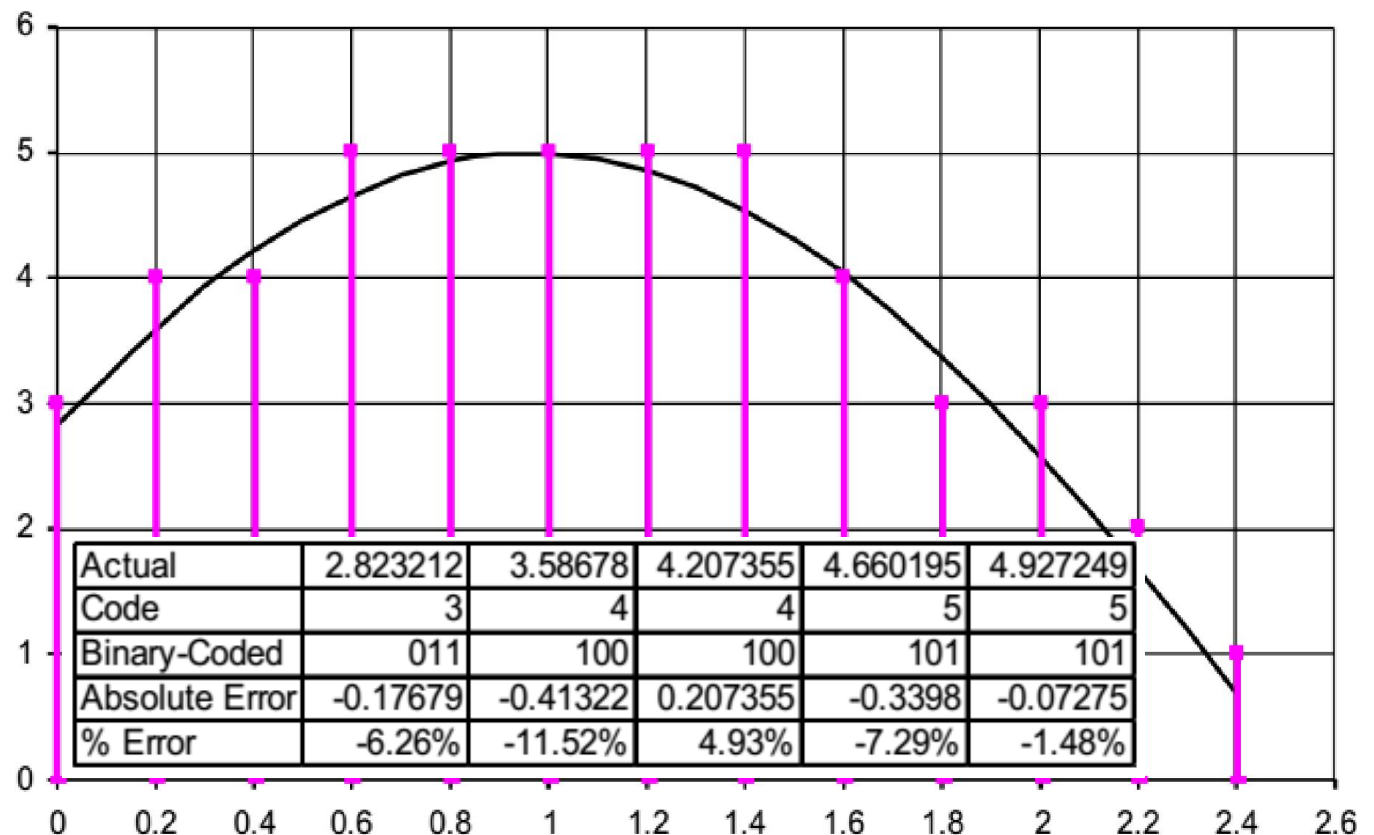
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Analog to Digital Conversion of Message/Signal

- Problem: quantization error is unavoidable

Quantization error can be minimized increasing L

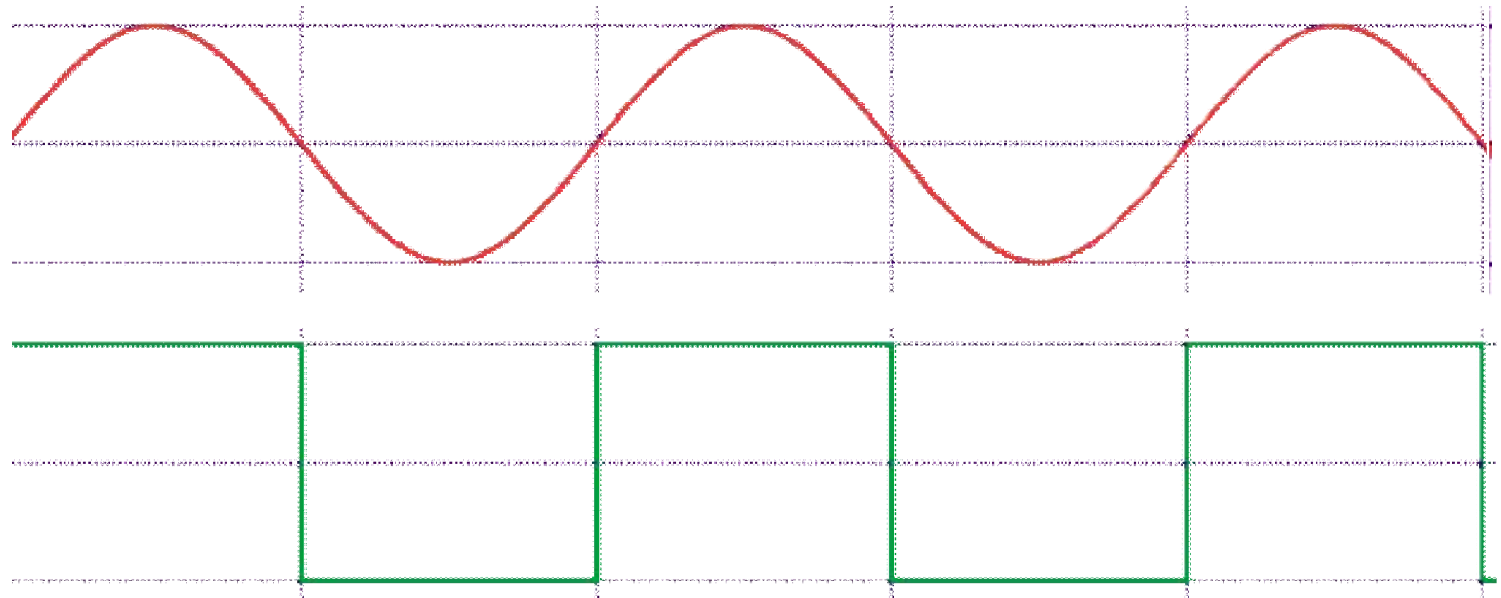


Representation of Digital Signal: Pulse Coded Modulation

- Assume, No. of quantization level, $L = 2$

Representation of Digital Signal: Pulse Coded Modulation

- Assume, No. of quantization level, $L = 2$
 - Easy to represent or transmit



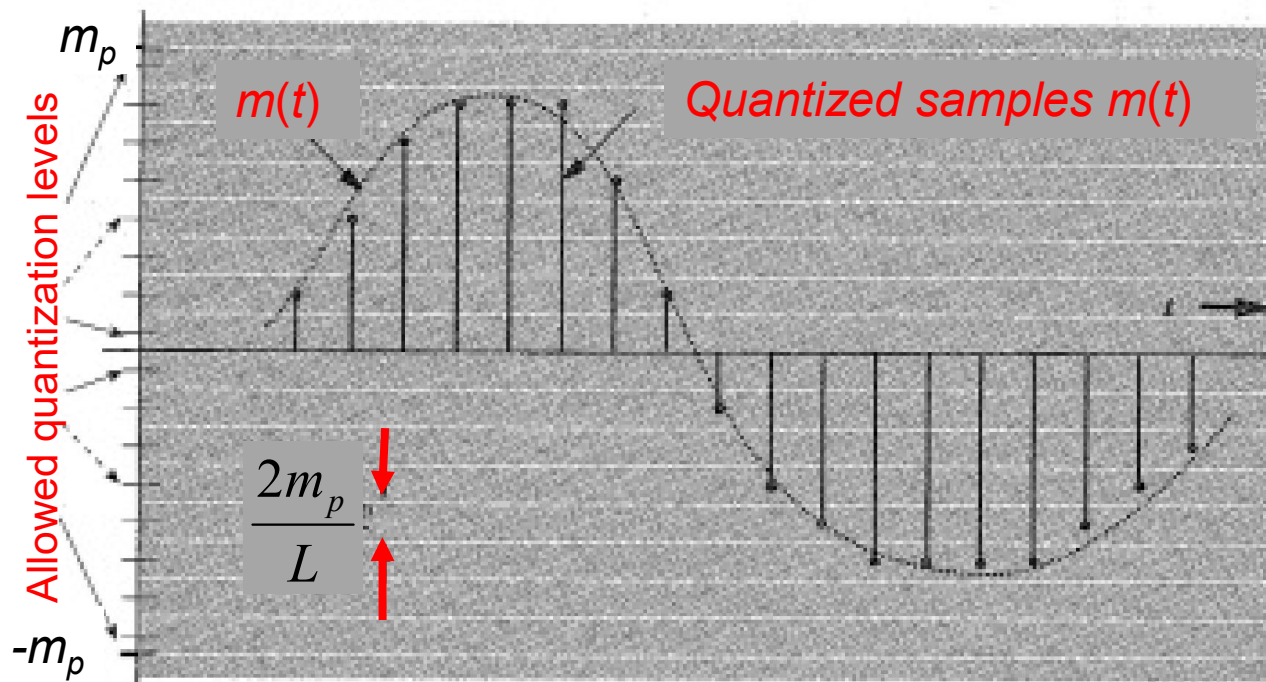
Representation of Digital Signal: Pulse Coded Modulation

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Representation of Digital Signal: Pulse Coded Modulation

- If No. of quantization level, $L \gg 2$

Each sample is represented by one of L levels

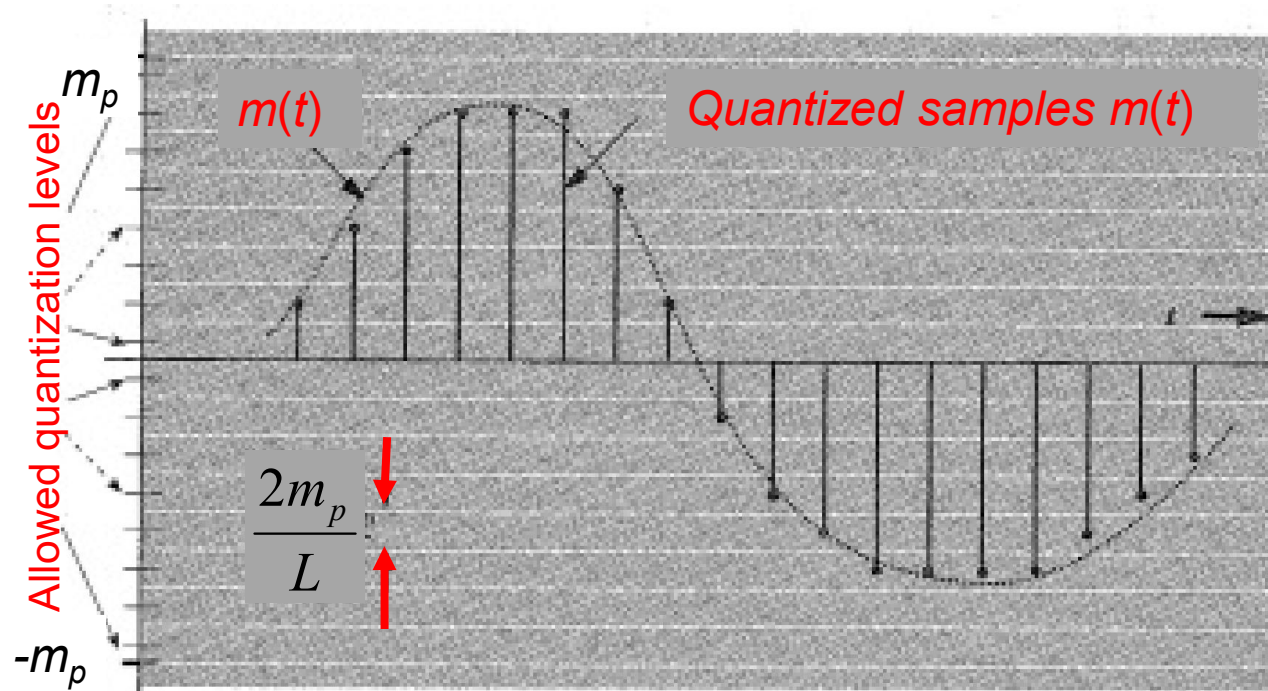


Representation of Digital Signal: Pulse Coded Modulation

- If No. of quantization level, $L \gg 2$
 - Solution is PCM
 - Each quantized value is represented by a *sequence of binary pulses*.

Representation of Digital Signal: Pulse Coded Modulation

- Assume, $L = 16$

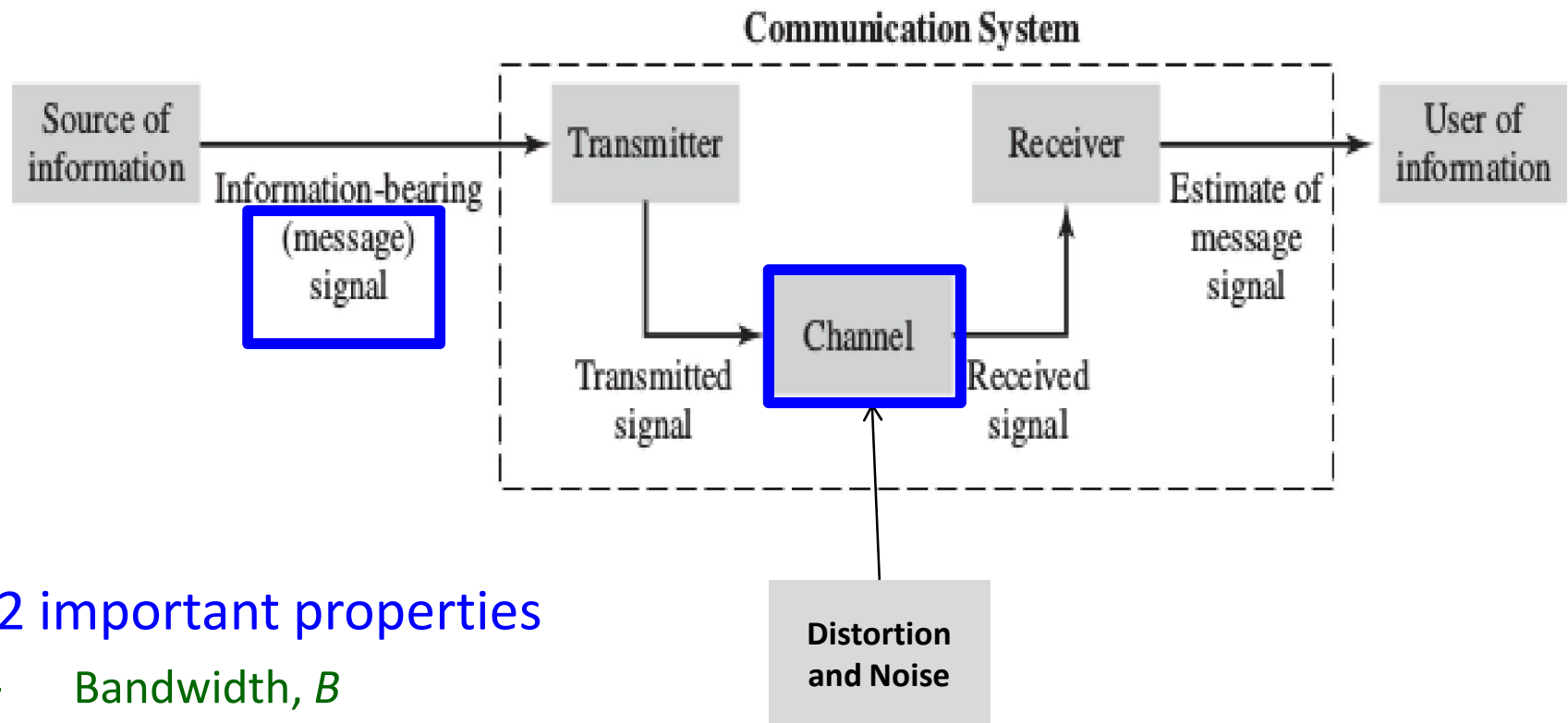


Representation of Digital Signal: Pulse Coded Modulation

- Assume, $L = 16$
 - Each quantized value is represented by a *sequence of FOUR binary pulses*.

Digit	Binary equivalent	Pulse code waveform
0	0000	
1	0001	
2	0010	
3	0011	
4	0100	
5	0101	
6	0110	
7	0111	
8	1000	
9	1001	
10	1010	
11	1011	
12	1100	
13	1101	
14	1110	
15	1111	

Channel & signal Characteristics



- 2 important properties
 - Bandwidth, B
 - Signal power, P_s

Channel & signal Characteristics

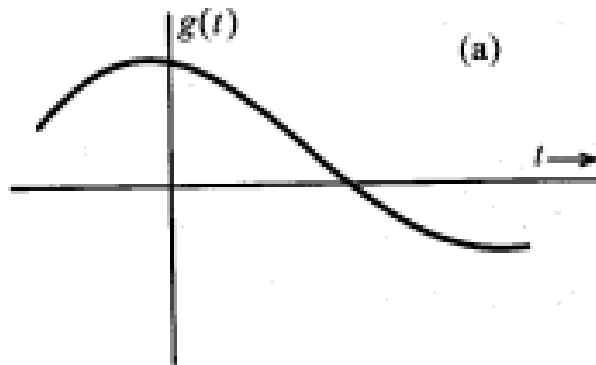
- Bandwidth
 - Channel Bandwidth
 - Signal bandwidth

Channel & signal Characteristics

- Bandwidth
 - Channel Bandwidth
 - Range of frequencies that a channel can transmit
 - Ex: if transmit 0-5 kHz frequencies, Channel bandwidth $B = 5$ kHz
 - Signal bandwidth

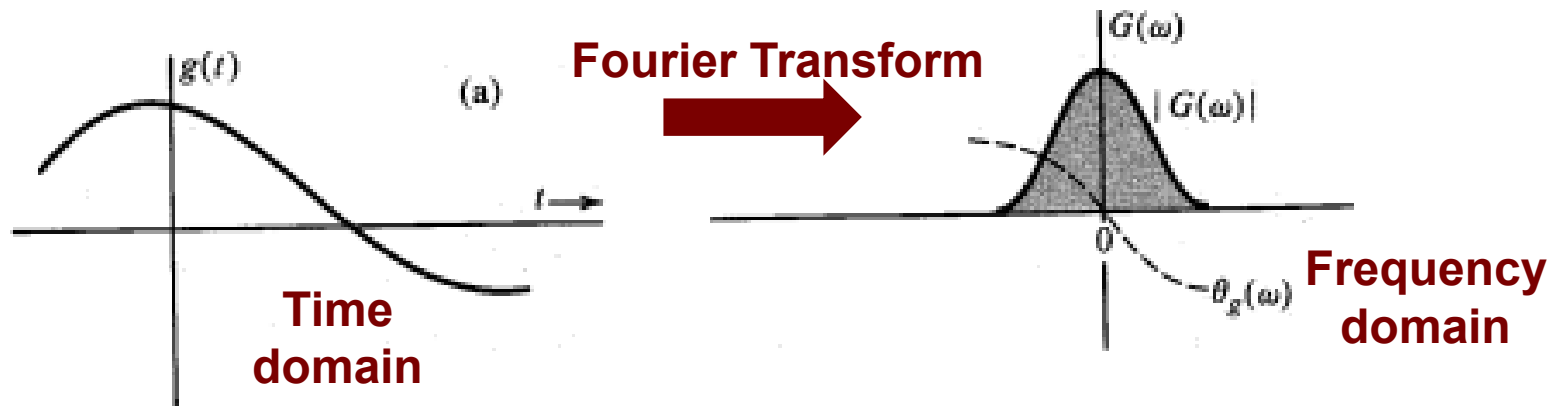
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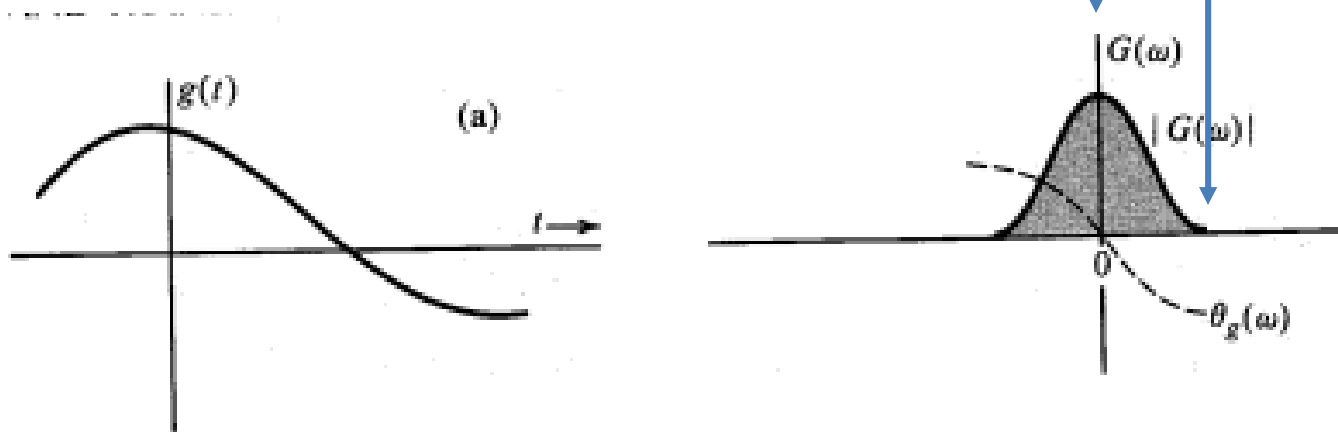
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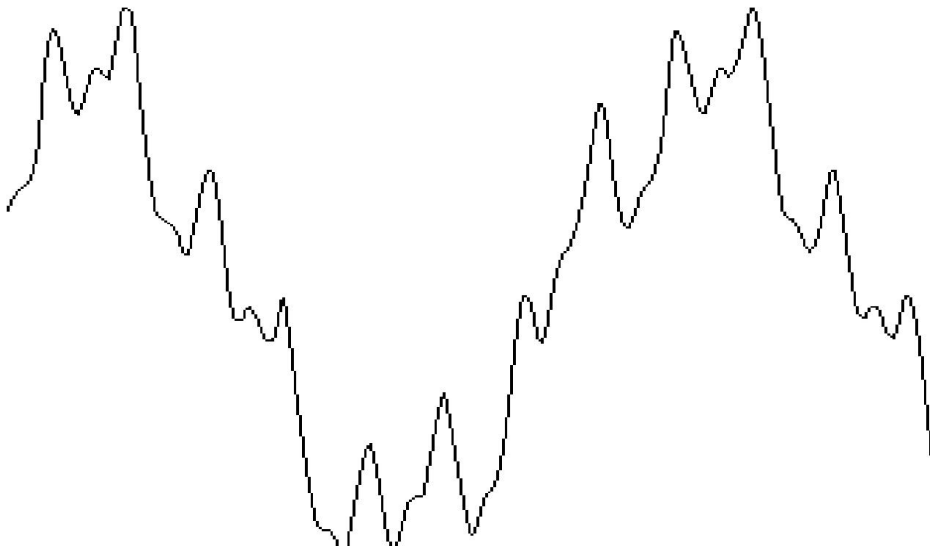


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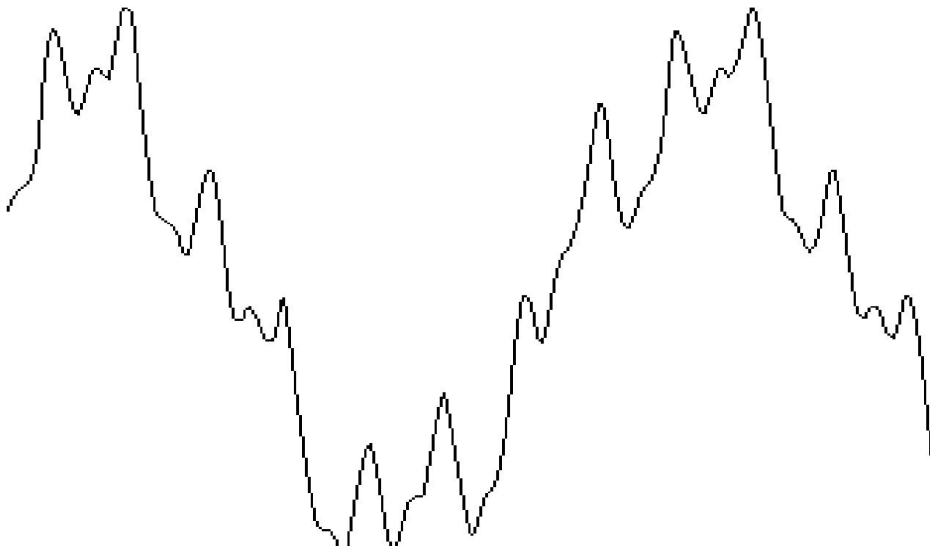


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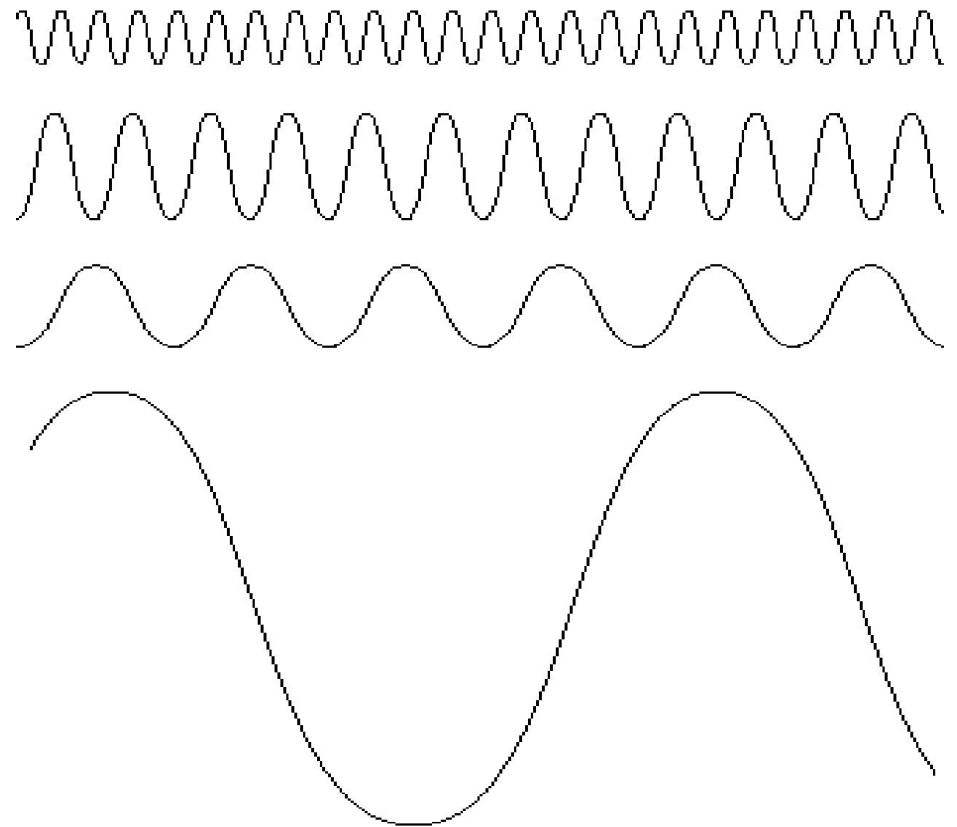


A complex signal with
multiple frequencies

Channel & signal Characteristics



A complex signal with
multiple frequencies



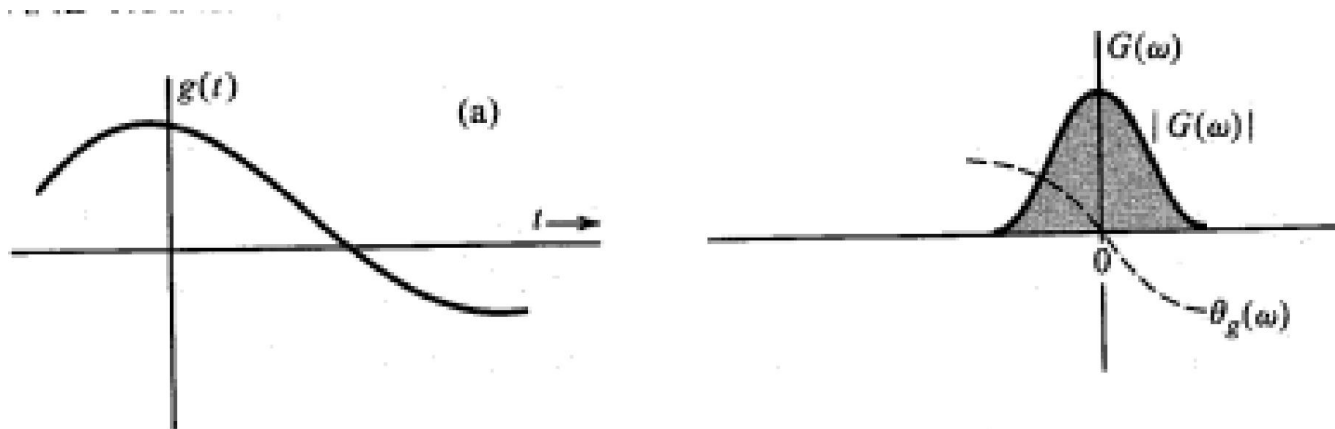
Individual signals each with
a single frequency

Channel & signal Characteristics

- Frequency
 - Change in signal values
 - Faster change in values means higher frequencies
 - High frequency signals
 - Sports/battle scene
 - Low frequency signals
 - News/sleeping animal videos

Channel & signal Characteristics

- Frequency
 - Compressing in time, increases frequency, means higher channel Bandwidth



Channel & signal Characteristics

- Frequency
 - Compressing in time, increases frequency, means higher channel Bandwidth

